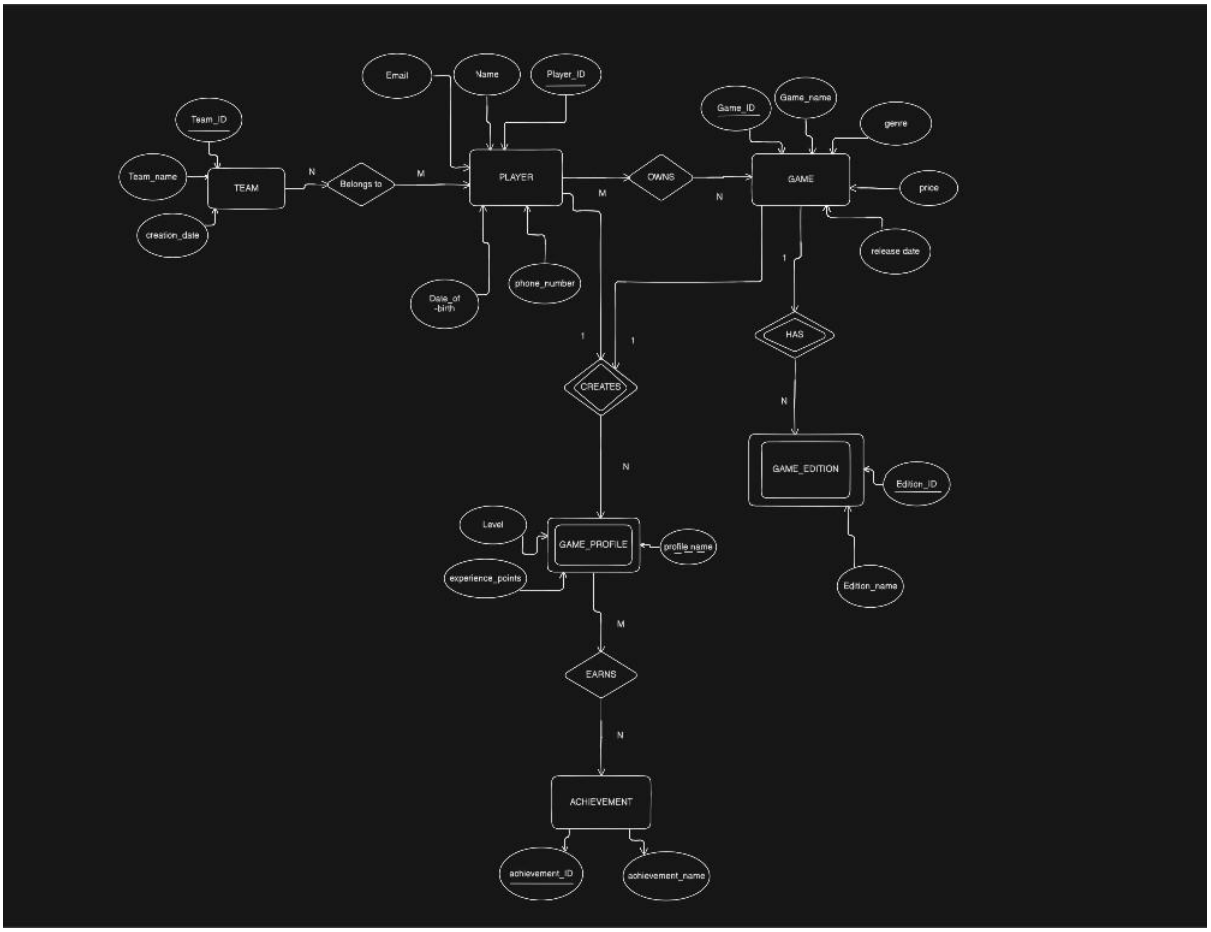
ER DIAGRAM – ONLINE GAMING PLATFORM

Name: Bhavya Sahithi G
USN: 1RUA25BCA0014

Course: BCA



1. ENTITIES AND ATTRIBUTES

PLAYER

- **Player_ID** – Primary Key
- Name
- Email
- Date_of_Birth
- Phone_Number

GAME

- **Game_ID** – Primary Key
- Game_Name
- Genre
- Release_Date
- Price

GAME_EDITION – Weak Entity

- **Edition_ID** – Partial Key
- Edition_Name

GAME_PROFILE – Weak Entity

- **Profile_Name** – Partial Key
- Level
- Experience_Points

ACHIEVEMENT

- **Achievement_Name** – Primary Key
- Description

TEAM

- **Team_ID** – Primary Key
- Team_Name
- Creation_Date

2. RELATIONSHIPS AND CARDINALITIES

OWNS

PLAYER — OWNS — GAME

Cardinality: **M:N**

A player can own multiple games, and a game can be owned by many players.

HAS

GAME — HAS— GAME_EDITION

Cardinality: **1:N**

This is an **identifying relationship** because GAME_EDITION cannot exist independently without GAME.

CREATES

PLAYER —CREATES — GAME_PROFILE

Cardinality: **1:N**

This is an **identifying relationship** because GAME_PROFILE cannot exist independently without PLAYER.

CREATES

GAME — CREATES— GAME_PROFILE

Cardinality: **1:N**

This is an **identifying relationship** because GAME_PROFILE cannot exist independently without GAME.

EARNS

GAME_PROFILE — EARNS— ACHIEVEMENT

Cardinality: **M:N**

A game profile can have several achievements, and an achievement can be associated with multiple profiles.

EARNS

PLAYER — EARNS — ACHIEVEMENT

Cardinality: **M:N**

A player may earn many achievements, and an achievement may be earned by many players.

BELONGS_TO

PLAYER — BELONGS_TO — TEAM

Cardinality: **M:N**

A player may belong to multiple teams, and a team can have multiple players.

3. WEAK ENTITY DETAILS

Weak Entity	Owner Entity	Partial Key	Identifying Relationship
GAME_EDITION	GAME	Edition_ID	HAS
GAME_PROFILE	PLAYER and GAME	Profile_Name	CREATES

4. EXPLANATION OF WEAK ENTITIES

GAME_EDITION

GAME_EDITION is a weak entity because it cannot exist independently without a GAME. Its Edition_ID is unique only within a particular game and therefore cannot uniquely identify an edition by itself. GAME is the owner entity, Edition_ID is the partial key, and HAS_EDITION is the identifying relationship. Therefore, an edition is identified using **Game_ID + Edition_ID**.

GAME_PROFILE

GAME_PROFILE is a weak entity because it cannot exist independently without its associated PLAYER and GAME. Profile_Name alone cannot uniquely identify a profile. PLAYER and GAME are the owner entities, Profile_Name is the partial key, and HAS_PROFILE and FOR_GAME are the identifying relationships. Therefore, a profile is identified using **Player_ID + Game_ID + Profile_Name**.

5. CHEN NOTATION USED

- **Single Rectangle** → Strong Entity
- **Double Rectangle** → Weak Entity
- **Single Diamond** → Relationship
- **Double Diamond** → Identifying Relationship
- **Oval** → Attribute
- **Underlined Attribute** → Primary Key
- **Dashed-Underlined Attribute** → Partial Key
- **M:N** → Many-to-Many Relationship
- **1:N** → One-to-Many Relationship

6. CONCLUSION

The ER diagram represents the Online Gaming Platform by identifying its entities, attributes, relationships, cardinalities, and weak entities. The diagram clearly shows the dependencies of GAME_EDITION on GAME and GAME_PROFILE on PLAYER and GAME using standard Chen notation.

